AI Factory

Evidence-Based Manufacturing Assistant

Guided quality investigation for factory engineers

Week 3: Users and Data | Week 4: Objectives and HAI Principle Ladder
Track: AI | Submission draft for team review

Personas are fictional. Dataset quantities are proposed targets until files are created. Prototype is not connected to a real factory database.

Project Overview

- Helps factory teams investigate recurring quality defects using inspection, production, maintenance and supplier/material records
- Reduces guesswork, keeps the user in control
- Separates **Observed Facts**, **Hypotheses**, **Missing Data**; shows source and time range; suggests a next verification step instead of presenting unverified causes as fact
- **Primary users:** QC engineers investigating recurring defects. **Secondary:** production engineers, maintenance teams, production managers
- **Scope boundary:** all examples and records are synthetic/illustrative; no real factory, employee or supplier data collected

1. Three Fictional Personas

	A: Mariam Hassan (below-typical)	B: Omar Nabil (typical)	C: Salma Farouk (advanced)
Role	QC engineer, medium packaging factory; 29, Egyptian; Arabic first, understands English labels	Production engineer coordinating with QC and maintenance; 35, Jordanian	Senior Quality Engineer / CI analyst; 41, Sudanese
Tech / AI	Spreadsheets and messaging only; novice AI user, may over-trust fluent answers	Spreadsheets, dashboards; novice-to-intermediate, checks claims	Dashboards, filters; regular AI user, knows answers can be wrong
Goal	Understand a defect rise in a period; decide what to verify	Check if production changes and defect rise share a window	Compare defects across batches, shifts, periods with traceability
Pain points	Time pressure, mixed formats, jargon, too many questions	Needs shareable summaries; formats differ; source dates omitted	Over-simplified answers hide assumptions
Accessibility	Plain language, readable text, headings, bilingual labels	Scannable layout, not colour-only	Keyboard controls, structured tables, not colour-only
Needs	Essential questions only; Fact/Hypothesis/Missing labels; source, time; one next step	Short summary, traceable evidence, uncertainty, verification step	Source traceability, filters, assumptions, missing-data warnings, control
Design	Concise, explain terms, uncertainty visible	Show source, window, sample size; correlation vs causation	Deep inspection without forced long flow

2. Bias-Check Notes

Issue found	Why harmful	Safeguard
Culture treated as predictor of technical ability	Stereotype instead of individual context	Technical level = software experience and confidence only
Vague accessibility need	Does not influence design	Map to readable text, keyboard support, headings, non-colour cues
Fluent AI sounds too certain	Hypothesis mistaken for cause	Labels: Observed Fact, Hypothesis, Missing Data
Higher defects in one shift	Volume, mix, coverage, missing records vary	Denominators, matching windows, neutral wording, no causation
Supplier judged on small sample	Sample size and product differences distort	Show sample size, tentative wording, like-for-like
Novices excluded by jargon	Misunderstand or abandon	Plain language; clarify only when needed
Advanced users blocked	Cannot inspect evidence	Optional detail and filters

Rule: personas are fictional design aids. Before user testing, review language with team members from different backgrounds; check accessibility is testable.

3. Data Source (synthetic, team-created)

Four proposed files: ~120 inspection records, 30 shift summaries, 25 maintenance events, 20 supplier/material-lot records (planning targets, not collected; replace with actual counts in final report). Fictional IDs, timestamps, values; no real names or confidential data. Generation method and permitted project use documented. If real/public data is added: document source, licence/permission, permitted uses, date, restrictions. Checks: missing values, duplicates, inconsistent timestamps, mismatched IDs; missing info stays visible. All outputs labelled synthetic/illustrative.

File	Example fields	Purpose	Limitation
inspection_records.csv	record_id, timestamp, product_id, batch_id, defect_type, inspected_qty, defect_qty	Defect patterns, compare batches/periods	Bad counts distort rates
shift_production.csv	record_id, date, shift_id, product_id, batch_id, units_produced, downtime_minutes	Compare production and quality	Overlap is not cause
maintenance_events.csv	event_id, timestamp, machine_id, event_type, status, notes	Check relevant events	Nearby event is not proof
supplier_lots.csv	lot_id, supplier_code, material_code, delivery_date, inspection_status	Material-lot patterns	Small samples mislead

Data Quality Plan and Week 3 Prompt Log

- Consistent timestamps, unique IDs, shared product/batch/machine identifiers
- Check duplicates, missing values, impossible counts, unmatched IDs
- Preserve original files; document each transformation; never fill missing values silently
- Matching time windows and denominators; label everything synthetic; claim no licence unless verified

Appendix A W3-1 Persona Draft (ChatGPT): prompt asked for 3 fictional personas (below-typical, typical, advanced; 2+ novice AI users; 1+ accessibility need; varied backgrounds; no stereotypes). Changes: separate culture from ability, concrete accessibility requirements, all details fictional.

W3-2 Data and Bias Plan (Gemini): prompt asked for a synthetic four-file plan, sample sizes, fields, risks (shift/worker blame, supplier bias, missing data, correlation vs causation), quantities labelled proposed. Changes: targets marked, licence claims limited to verifiable, safeguards added.

Accuracy note: the team must make tool, date, exact prompt, output and edits match actual use.

Appendix A W3-1: Persona Draft

Tool and purpose

Tool: ChatGPT. Draft fictional personas that cover required technical levels and accessibility needs.

Exact prompt to record if used

“Using this project context, AI Factory is an evidence-based manufacturing assistant for quality investigations. Draft three fictional user personas: one below-typical technical level, one typical, and one advanced. Across the three, include at least two novice AI users, at least one accessibility need, and varied cultural backgrounds. For each persona include role, background, technical level, AI experience, goals, context, pain points, accessibility needs, and design implications. Do not use real personal data and do not rely on cultural stereotypes.”

Output summary

Three fictional persona drafts with different technical confidence levels, AI experience, goals, pain points, and interface needs.

Review and changes

Keep cultural background separate from technical ability; convert accessibility needs into concrete design requirements; ensure two personas are novice or novice-to-intermediate AI users; keep all names and details fictional.

Appendix A W3-2: Data and Bias Plan

Tool and purpose

Tool: ChatGPT. Propose a safe synthetic data plan and identify bias risks.

Exact prompt to record if used

“For AI Factory, propose a synthetic prototype data plan for inspection records, shift-production summaries, maintenance events, and supplier/material-lot records. Suggest manageable sample sizes and example fields. Identify risks including shift blame, worker blame, supplier bias, missing data, and confusing correlation with causation. Clearly label all quantities as proposed rather than collected. Do not invent a real dataset, source, license, or permission.”

Output summary

A proposed four-file synthetic data plan, target record counts, data-quality checks, and a bias-risk checklist.

Review and changes

Quantities explicitly marked as targets; source/license claims limited to what the team can verify; safeguards added against unsupported causal claims and unfair blame. Accuracy note: tool, date, prompt, output and edits must match actual use.

5. Project Plan v1: Goal, Scope, Workflow

Goal: prototype that retrieves synthetic records, summarizes supported patterns, shows evidence and uncertainty, suggests a next verification step. **In scope:** synthetic CSVs; text interface; guided flow for ambiguous questions; concise answers for simple ones; evidence display; Fact/Hypothesis/Missing labels; user-selected next steps. **Out:** real factory database; machine control; automatic disciplinary/supplier decisions; causation claims; real personal data.

1. User describes a defect or asks a question
2. System checks product, batch, machine, time period are clear
3. Asks one concise clarification only if missing info affects reliability
4. Searches approved synthetic source and analyzes matching records
5. Presents Observed Facts (source, time range), Hypotheses (supporting/limiting evidence), Missing Data
6. Suggests verification steps; user chooses
7. User reviews, gives feedback or corrects context, finishes or restarts

Tool Choices and Grounding / Safety Rules

Layer	v1 choice	Reason
Model	Hosted LLM API (subject to access/budget)	Low setup; no real personal data; fall back to approved local model, documented
Backend / UI	Python + FastAPI / Streamlit	Team skills; quick forms, tables, evidence panels
Knowledge	Version-controlled synthetic CSVs + data dictionary	Bounded, inspectable, reproducible
Retrieval	pandas filters, deterministic calculations before LLM	Code computes counts/rates; LLM explains only
Grounding	Retrieve rows, attach file/record IDs and time range, answer only from rows	Missing evidence must be stated
Versioning	Git, saved CSVs, repeatable test questions	Repeatable demos

Confirm which API/model is actually available; this is a proposed architecture, not completed. **Rules:** every factual claim backed by rows or labelled assumption; show source and time range; separate observed from interpreted; no causation from correlation; disclose too few records, mismatched windows, missing fields; never invent maintenance, inspection or supplier records; label demo outputs synthetic.

6. Journey Maps: Start, Ask, Understand, Act/Fix, Finish

Stage	A: Mariam	B: Omar	C: Salma
Start	Unsure what to prepare. Inform: purpose, example questions	Needs quick shareable answer. Inform: scope, limits; evidence not verdict	Wizard feels restrictive. Inform: scope plus detail option
Ask	"Why did defects increase yesterday?" lacks product/time. Control: one clarification only if needed	Departments use different windows. Control: choose range/product	Needs precise controls. Control: filters, editable scope
Understand	Jargon; over-trust. Explain: plain summary, source, labels	Coincidence vs causation. Explain: aligned ranges, sample sizes	Distrusts model reasoning. Explain: evidence rows, calculation, assumptions
Act / fix	Unsure which check helps. Offer practical next steps	Generic advice hard to action. Concrete options	Needs flexibility. User picks next analysis
Finish	Helpful / not helpful, optional comment	Traceability: source IDs plus feedback	Helpful plus reason category/comment

Success signals: A identifies fact vs hypothesis, one-action feedback. B states what evidence does/does not support; summary traceable. C traces claims to source, changes direction without restart, feedback identifies a correction.

Journey Map A: Mariam (below-typical confidence)

Stage	User action / question	Likely pain point	Principle / system response	Success signal
Start	Opens AI Factory to investigate a packaging defect	Unsure what information to prepare	Inform: short purpose statement and example questions	Can identify what the assistant helps with
Ask	Types: “Why did defects increase yesterday?”	Question lacks product or exact time range	Allow control: one short clarification only if needed; user chooses period/product	Completes the question without confusion
Understand	Reads facts, hypotheses, and missing data	Technical language; may over-trust a fluent answer	Explain: plain-language summary, source/time range, clear uncertainty labels	Correctly identifies fact vs hypothesis
Act / fix	Chooses to compare defect rates for a product and shift	May not know which check is useful	Allow control: practical next verification steps, not a forced decision	Selects a relevant next step
Finish	Reviews summary, optionally gives feedback	May not know whether answer was useful	Enable feedback: simple helpful/not helpful and optional comment	Feedback submitted in one short action

Journey Map B: Omar (typical confidence)

Stage	User action / question	Likely pain point	Principle / system response	Success signal
Start	Opens the assistant before a cross-team production meeting	Needs an answer that can be shared quickly	Inform: scope and source limitations	Understands the result is evidence support, not a final verdict
Ask	Asks whether a production change coincided with a defect increase	Different departments may use different time windows	Allow control: choose date range/product; clarify only if needed	Question has a clear scope
Understand	Checks summary and supporting records	May confuse coincidence with causation	Explain: aligned time ranges, sample sizes, limitations	Can state what evidence supports and does not support
Act / fix	Shares a verification task with QC or maintenance	Generic recommendation hard to action	Allow control: concrete verification options, user selects	Identifies a specific follow-up check
Finish	Copies the evidence summary into team notes	May need traceability later	Inform / Enable feedback: source IDs and feedback option	Summary stays traceable to records

Journey Map C: Salma (advanced confidence)

Stage	User action / question	Likely pain point	Principle / system response	Success signal
Start	Opens a recurring-defect investigation	A simple wizard may feel restrictive	Inform: scope plus option to inspect details	Can choose concise or detailed investigation
Ask	Filters by batch, machine, and date range	Needs precise controls, not repeated questions	Allow control: expose filters and editable scope	Selected filters match the intended investigation
Understand	Inspects source rows and computed patterns	May distrust unsupported model reasoning	Explain: evidence rows, calculation basis, assumptions, missing fields	Can trace a claim back to its source
Act / fix	Chooses a comparison or verification step	Needs flexibility to pursue another hypothesis	Allow control: user selects next analysis; workflow not locked	Changes direction without restarting
Finish	Reviews findings, flags a misleading or incomplete answer	Feedback may be too shallow to diagnose	Enable feedback: helpful/not helpful plus reason category/comment	Feedback identifies a correction or improvement

7. Measurable Objectives (proposed targets, not results)

Metric	Definition	Target	Measure
M1 Evidence-grounded rate	Factual claims supported by rows or labelled unavailable	≥90%, 20 questions	Manual audit vs CSVs
M2 Fact/Hypothesis/Missing labelling	Answers correctly separating the three	≥95%, 20 questions	Checklist
M3 Task completion	Users finish and pick next step unaided	≥80%, 5–8 users	Observation
M4 Time to first useful answer	Median time, clear question to labelled answer	≤60 s	Timestamps, excl. setup
M5 Appropriate clarification	Asks when ambiguous, not when clear	≥85%, 20 prompts	Labelled test set
M6 Understanding of uncertainty	Novices identify one fact and one hypothesis	≥80%	Post-task check
M7 Feedback completion	Users find and submit feedback	≥90%, 5–8 users	Usability task

8. HAI Principle Ladder v1

Level	Principle	Interface behaviour	Linked target
1 Inform	Say what the assistant does and can/cannot know	Scope, synthetic notice, source/time range, missing-data notice	M1 ≥90%
2 Explain	Make reasons and limits understandable	Fact / Hypothesis / Missing Data; supporting records, limitations	M2 ≥95%; M6 ≥80%
3 Allow control	Choices stay with the human	Set product, batch, machine, time; necessary questions only; next-step choices	M5 ≥85%; M3 ≥80%
4 Enable feedback	Report incorrect, incomplete, confusing answers	Helpful/not helpful plus reason; review test failures	M7 ≥90%

W4-1 Objectives and Ladder (ChatGPT): 4+ measurable objectives (AI quality, UX, principle-linked) with targets and methods, mapped to the ladder; targets not described as achieved. **W4-2 Journeys and Architecture:** journey maps with pain point, principle, success signal; feasible architecture; proposal distinguished from implementation; added source IDs, time ranges, safeguards. Verify logs against actual use.

Appendix A W4-1: Objectives and HAI Ladder

Tool and purpose

Tool: ChatGPT. Draft measurable prototype objectives and map them to the four HAI ladder levels.

Exact prompt to record if used

“For AI Factory, an evidence-based manufacturing assistant using synthetic inspection, production, maintenance, and supplier/material records, propose at least four measurable v1 objectives. Include at least one AI-quality metric, one user-experience metric, and one metric tied to Inform, Explain, Allow control, or Enable feedback. Give a numeric target and a practical measurement method for each. Build a four-level HAI Principle Ladder using Inform, Explain, Allow control, and Enable feedback, and link each level to measurable objectives. Do not describe targets as achieved results.”

Output summary

A proposed metric set with numeric targets and measurement methods, plus a four-level ladder linked to those objectives.

Review and changes

Targets identified as proposed; metrics include AI quality, UX, clarification/control, uncertainty understanding, and feedback. Actual outcomes reported only after testing.

Appendix A W4-2: Journey Maps and Tool Architecture

Tool and purpose

Tool: ChatGPT. Draft persona-specific journeys and a grounded v1 architecture.

Exact prompt to record if used

“For the three fictional AI Factory personas (below-typical, typical, advanced), draft journey maps covering Start, Ask, Understand the answer, Act/fix, and Finish. At each stage include a pain point, a relevant HAI principle, and a success signal. Recommend a feasible student prototype architecture with model, backend, interface, synthetic knowledge source, retrieval/grounding, and safety rules. Clearly distinguish proposed design from implemented functionality.”

Output summary

Three persona journey maps and a proposed architecture using Python, a lightweight interface, synthetic CSV files, deterministic retrieval/calculation, and grounded model summaries.

Review and changes

Architecture clarified as a proposal until implemented; source identifiers and time ranges required; safeguards added against unsupported causation and silent guessing of missing data. Verify the log against the team's actual use.

10. Demo Readiness Checklist

- Explain the three personas and how needs change the design
- Show the synthetic CSVs, or state they are planned
- Run the prototype only if implemented; claim no working integration otherwise
- Demo one clear and one ambiguous question; clarification only when necessary
- Show Observed Fact, Hypothesis, Missing Data, source, time range, next step
- Show choosing a next step and submitting feedback
- Explain each metric, target, measurement, and each HAI level with its metric
- Keep prompt logs accurate: exact prompts, actual tool, output summary, changes, reasons
- Send no real personal data, confidential records or real screenshots to an AI tool